

ASSIGNMENT 03
Due date: Friday, 31 July 2026
Total Marks: 20

ONLY FOR YEAR MODULE

This assignment covers **chapters 1 & 2** of the prescribed book as well as the study guide, **Note that this assignment will close on Thursday, 31 July 2026 at 9:00 PM**

Question 1: 20 Marks

A square matrix Q is called **orthogonal** if its transpose is equal to its inverse, i.e., $Q^T = Q^{-1}$, which is equivalent to the condition $Q^T Q = I$.

- a) Prove that if Q is any $n \times n$ orthogonal matrix, then its determinant must be either 1 or -1 .
- b) For any non-zero column vector $\vec{v} \in \mathbb{R}^n$, the **Householder matrix** is defined as:

$$H = I - 2 \frac{\vec{v}\vec{v}^T}{\|\vec{v}\|^2}$$

Prove that any Householder matrix H is both symmetric ($H^T = H$) and orthogonal ($H^T H = I$).

- c) Let $\vec{v} = \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix}$. Construct the specific Householder matrix H corresponding to this vector. Then, for the vector $\vec{x} = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}$, compute the transformed vector $\vec{y} = H\vec{x}$. Describe the geometric relationship between $\vec{v}$, $\vec{x}$, and $\vec{y}$.